CSDF-5: Honeypot

Program (honeypot.py): low-interaction honeypot

```
import socket, threading, time, datetime, sys
PORTS = {2121: b"220 FTP Server ready\r\n", 2323: b"Login: ", 8081: b"HTTP/1.1 200 OK\r\nServer:
    Apache/2.2\r\n\r\n"}
LOG = "honeypot.log"

def log(msg):
    line = f"{datetime.datetime.now():%Y-%m-%d %H:%M:%S} {msg}"
    print(line); open(LOG, "a").write(line + "\n")

def handle(conn, addr, port):
    log(f"[CONNECT] {addr[0]}:{addr[1]} -> port {port}")
    conn.sendall(PORTS[port]); conn.settimeout(2)
    try:
        data = conn.recv(1024)
        if data: log(f"[DATA]    {addr[0]} sent {data.strip()!r}")
    except socket.timeout: pass
    conn.close(); log(f"[CLOSED]  {addr[0]}:{addr[1]}")

def listen(port):
    s = socket.socket(); s.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    s.bind(("127.0.0.1", port)); s.listen(5)
    while True:
        c, a = s.accept(); threading.Thread(target=handle, args=(c, a, port), daemon=True).start()

if __name__ == "__main__":
    for p in PORTS: threading.Thread(target=listen, args=(p,), daemon=True).start()
    log(f"Honeypot listening on ports {list(PORTS)}")
    time.sleep(float(sys.argv[1]) if len(sys.argv) > 1 else 3600)
```

Output: log after 4 test connections

```
2026-10-28 06:16:45 Honeypot listening on ports [2121, 2323, 8081]
2026-10-28 06:16:46 [CONNECT] 127.0.0.1:59386 -> port 2121
2026-10-28 06:16:46 [DATA]    127.0.0.1 sent b'USER admin'
2026-10-28 06:16:46 [CLOSED]  127.0.0.1:59386
2026-10-28 06:16:46 [CONNECT] 127.0.0.1:38276 -> port 2323
2026-10-28 06:16:46 [DATA]    127.0.0.1 sent b'root'
2026-10-28 06:16:46 [CLOSED]  127.0.0.1:38276
2026-10-28 06:16:47 [CONNECT] 127.0.0.1:39972 -> port 8081
2026-10-28 06:16:47 [DATA]    127.0.0.1 sent b'GET /admin HTTP/1.1'
2026-10-28 06:16:47 [CLOSED]  127.0.0.1:39972
2026-10-28 06:16:47 [CONNECT] 127.0.0.1:59394 -> port 2121
2026-10-28 06:16:47 [DATA]    127.0.0.1 sent b'USER test'
2026-10-28 06:16:47 [CLOSED]  127.0.0.1:59394
```